

Aim: Simulation of code optimization Techniques. Constant Propagation

To implement a simple program to demonstrate Constant Propagation by comparing code with and without constant replacement.

Algorithm

1. Start the program and initialize pi using `floor((22.0/7.0)*100)/100` for demonstration.
2. Display a menu with three options:
 - Option 1: Calculate perimeter and area of a circle using pi from memory (without constant propagation).
 - Option 2: Replace pi with 3.14 in expressions (simulating constant propagation).
 - Option 3: Exit.
3. For both options 1 and 2:
 - Prompt the user to enter the radius.
 - Compute and print the perimeter and area of the circle.
4. Repeat until the user chooses to exit.
5. End the program.

Program Code

```
#include <stdio.h>
#include <math.h>

int main ()
{
    int c;
    float r, pi = floor((22.0 / 7.0)*100)/100;
    do
    {
        printf ("\nMENU");
        printf ("\n1. Find perimeter and area of circle without constant propagation");
        printf ("\n2. Find perimeter and area of circle with constant propagation");
        printf ("\n3. Exit");
        printf ("\nEnter your Option : ");
```

```

scanf ("%d", &c);
switch (c)
{
case 1:
    printf ("Enter the radius of circle : ");
    scanf ("%f", &r);
    printf ("Perimeter = %0.2f", 2 * pi * r);
    printf ("\nArea = %0.2f", pi * r * r);
    break;

case 2:
    printf ("Enter the radius of circle : ");
    scanf ("%f", &r);
    printf ("Perimeter = %0.2f", 2 * 3.14 * r);
    printf ("\nArea = %0.2f", 3.14 * r * r);
    break;
}
}
while (c != 3);
return 0;
}

```

Sample Output

MENU

1. Find perimeter and area of circle without constant propagation
2. Find perimeter and area of circle with constant propagation
3. Exit

Enter your Option : 1

Enter the radius of circle : 5

Perimeter = 31.42

Area = 78.50

Enter your Option : 2

Enter the radius of circle : 5

Perimeter = 31.40

Area = 78.50